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MILITARY SPECIFICATION

PUMPING ASSEMBLY, WATER, WHEEL MOUNTED, DIESEL-ENGINE-DRIVEN, SELF-PRIMING CENTRIFUGAL, 6-INCH, 600 GPM AT 350 FEET TOTAL HEAD

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a wheel-mounted, diesel-engine-driven, self-priming centrifugal pump, 6-inch for use in a potable water hoseline system.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-C-390 - Copper Alloy Castings.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-V-173	- Varnish, Moisture and Fungus Resistant (For Treatment of Communications Electronics and Associated Equipment).
MIL-P-514	- Plate, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-G-3859	- Grease Gun, Hand Operated, Lever, Push and Screw Type.
MIL-A-8421	- Air Transportability Requirements, General Specification for.
MIL-H-8794	- Hose, Rubber, Hydraulic, Fuel and Oil Resistant.
MIL-E-10062	- Engine, Preparation for Shipment and Storage of.
MIL-C-10387	- Coupling, Clamp, Pipe with Bolts and Synthetic Rubber Gasket for Grooved Pipe and Tube.
MIL-P-10603	- Pumps and Pumping Units, Centrifugal, Power Driven, for Water, Packaging of.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-P-15024	- Plate, Tags and Bands for Identification of Equipment.
MIL-P-15024/9	- Plate, Tags, and Bands, Aircraft Loading Data Plate.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-A-46153	- Antifreeze, Ethylene, Glyco, Inhibited, Heavy Duty, Single Package.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-E-52649	- Engine, Cold Starting Aids, Ether Fuel Primers.
MIL-C-62122	- Cable Assembly, Slave, Electrical Connectors, Plug and Receptacle.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.

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MIL-STD-130	- Identification Marking of U.S. Military Property.

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MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1366	- Packaging, Handling, Storage, and Transportation System Dimensional Constraints, Definition of.
MIL-STD-1410	- Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiels.
MS24532	- Voltmeter, Battery Generator, 28 Volt DC.
MS27022	- Coupling Half, Quick Disconnect, Cam-Locking Type, Male, External Pipe Thread, Type III.
MS27028	- Coupling Half, Quick Disconnect, Cam-Locking Type, Cap, Dust, Type IX.
MS35000	- Batteries, Storage, Lead Acid.
MS35643	- Gasket: Synthetic Rubber, Cap, Fuel Tank.
MS35644	- Filler Neck: Fuel Tank-Military Vehicles.
MS35645	- Cap, Fuel Tank: Tactical Vehicle, Fording.
MS52125	- Composite Light-Tail, Stop, Turn and Marker.
MS75020	- Connector, Plug, Electrical-12 Contact Intervehicular, 24 Volt Waterproof.

HANDBOOKS

MILITARY

MIL-HDBK-157	- Transportability Criteria.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

FOOD AND DRUG ADMINISTRATION (FDA)

Code of Federal Regulations, Title 21, Chapter 1, Part 177.

(The Code of Federal Regulations [CFR] and the Federal Register [FR] are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

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(Copies of specifications, standards, handbooks, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY (AWS)

D1.1 - Structural Welding Code - Steel.

(Application for copies should be addressed to the American Welding Society, Inc., 550 N.W. LeJeune Rd, P.O. Box 351040, Miami, FL 33135.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 48 - Gray Iron Castings, Spec for.
B 584 - Copper Alloy Sand Castings for General Applications, Spec for.
D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

HYDRAULIC INSTITUTE

Rating Standards and Test Code Standards.

(Application for copies should be addressed to the Hydraulic Institute, 122 East 42nd Street, New York, NY 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE J614 - Marking For Oil Dipsticks.
SAE J847 - Full Trailer Tow Bar Eye.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

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SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to South Coast Air Quality Managment District, 9150 East Flair Drive, El Monte, CA 91731.)

TIRE AND RIM ASSOCIATION, INC.

Year book.
Supplementary Service Data Book.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specifications sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The pump is a wheel-mounted, diesel-engine-driven, self-priming centrifugal pump. The pumping assembly has an operational point of 600 GPM at 350 feet total head when pumping liquids with a specific gravity of 1.0. The pumping assembly is equipped with automatic pump/engine controls.

3.2 First article. Unless otherwise specified (see 6.2), a pumping assembly shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of production pumping assemblies from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish pumping assemblies conforming to this specification.

3.3 Materials. The material shall be as specified herein. When a definite material is not specified, a material shall be used which will enable the pumping assembly to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guarantee of the acceptance of the finished product. The finished pumping assembly shall contain no materials or substances which might leak out or disintegrate and cause the water to become nonpotable. The material shall have no adverse effect on the health of personnel when used for its intended purpose. The lines and all surfaces that contact potable water shall conform to the Federal Food, Drug and Cosmetic, Food Additives, Chapter I, Title 21, Part

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177, Code of Federal Regulations for contact with food. The compounds listed in section 177.1020, 177.1030, 177.1040, 177.1050, and 177.1480 of the Federal Food, Drug and Cosmetic, Indirect Food Additives, Chapter I, Title 21, Code of Federal Regulations shall not be acceptable materials. Questions pertinent to this effort shall be referred by the procuring activity to the departmental medical service who will act as advisor to the procuring agency.

3.3.1 Material deterioration and control. The pumping assembly shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating enviornments to which the pumping assembly may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish, or treatment for use with components and subcomponents, and shall make information available to the contracting officer or designated representative upon request.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the pumping assembly may be newly fabricated from recovered materials to the maximum extent practicable, provided the pumping assembly produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the pumping assembly.

3.4 Manufacturer's product lines. The pump-engine combination and its components shall be the latest model(s) of the contractor (and any subcontractors). Product improvements incorporated in the manufacturer's production model are acceptable. The pumping assembly shall be equipped with all components necessary to enable the pumping assembly to function reliably and efficiently in sustained operation.

3.5 Design. The pumping assembly shall be designed to permit easy accessibility for maintenance and service in the field. The design shall be such as to prevent conditions hazardous to personnel or deleterious to equipment.

3.5.1 Ease of maintenance. All major assemblies and installed attachments shall be accessible for repair, replacement, and adjustment without the removal of other major assemblies and installed attachments not usually removed. Covers and plates which must be removed for component adjustment, repair, or replacement shall be equipped with quick-disconnect fasteners (no tools required, maximum one complete turn to release for rotary operated). The number and types of bolts and screws (only flat or pan head) shall be minimized; the same size

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fastener shall be used for all mounting applications. Various size fasteners shall be readily distinguishable. Drain outlets shall be located for accessibility. A maintenance schedule shall be furnished prior to the start of any testing. Design for maintainability shall be as specified in 3.9.

3.5.2 Reliability. The specified mean-time-between-failures shall be 200 hours when the pumping assembly is tested in accordance with 4.6.2.13.

3.6 Threads. Screw threads shall be in accordance with FED-STD-H28.

3.7 Environmental.

3.7.1 Operating temperature. The pumping assembly shall perform as specified herein in any ambient temperature from +125 to +32 °F.

3.7.2 Storage temperature. The pumping assembly shall not be damaged by storage in any ambient temperature from +155 to -65 °F.

3.8 Safety. All shafts or rotating parts of the pumping assembly shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All hot surfaces of the equipment, including exhaust pipes exposed to contact by personnel or which create a fire hazard, shall be fully guarded or insulated. Fuel tanks shall be located in a manner which will not allow spills or overflows to run onto the engine, exhaust, or electrical equipment. Adequate shielding may be used to attain the same result. Exhausts or discharges from the equipment shall be directed so that they do not endanger personnel or obstruct the view of the operator. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard.

3.9 Human factors engineering. The pumping assembly shall conform to the human factors engineering design criteria as specified in MIL-STD-1472. Special design emphasis shall be given, but not limited, General Requirements (4), Control/-
Display Integration (5.1), Visual Displays (5.2), Controls (5.4), Labeling (5.5), Anthropometry (5.6), Design for Maintainability (5.9), and Hazards and Safety (5.13) of MIL-STD-1472, as applicable.

3.10 Noise limits. The noise produced by the pumping assembly shall conform with MIL-STD-1474 requirements with the exception of MIL-STD-1474 5.2, 5.3, and 5.4 when tested in accordance with 4.6.2.8. The provisions of MIL-STD-1474, 4.3, and 4.4 shall be provided if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85dB(A) limit is obtained from the procuring activity. Hazard signs shall conform with MIL-STD-1474, 4.3 except the sign shall state "Hearing Protection Required Within (Specify) Feet". The Sign shall be readable at the distance indicated on the sign.

3.11 Pump.

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3.11.1 Performance. The pumping assembly shall be a 6-inch integrally self-priming (after case has been initially filled) centrifugal pump. The pumping assembly shall have a rated capacity of 600 gpm at 350 feet total head when pumping 1.0 specific gravity liquids.

3.11.1.1 Tilted position. The pumping assembly shall deliver its rated capacity when positioned at any angle from 0 to 15 degrees from the horizontal in any direction.

3.11.1.2 Priming. The pumping assembly shall prime and produce rated capacity within 2 minutes on a 10-foot static suction lift after initial filling of the pump casing.

3.11.2 Construction.

3.11.2.1 Pump case and volute. The pump case and volute shall be a one-piece gray iron casting conforming to ASTM A 48, class 30. The cutwater shall be a gray alloy casting conforming to ASTM A 48, class 30S, and shall be integrally cast into the pump case. The case shall have at least two mounting feet, provisions for complete pump drainage in 1 minute, and a 1-1/2-inch priming port with a connector check valve, coupling, and dust cap.

3.11.2.2 Impeller. The impeller shall be fabricated of bronze conforming to ASTM B 584, alloy C83600. The wear ring shall be copper alloy No. 836 per QQ-C-390.

3.11.2.3 Seals. The pump will be equipped with self-lubricating mechanical seals that permit operation of the pumping assembly in any operating condition specified herein.

3.11.2.4 Priming port. The pump shall be equipped with a 1-1/2 inch priming port fitted with a quick-disconnect, cam-locking type male coupling conforming to MS27022-9. The coupling shall have a cam-locking type dust cap conforming to MS27028-9 with a short length of chain securing it to the pump. A connector check valve shall be installed between the priming port coupling and the pump housing to prevent back flow from the pump casing.

3.11.2.5 Piping and fittings. All piping, valves, and fittings on the suction and discharge of the pump shall be flanged and fabricated from ASTM A 48, class 30, gray iron, except as specified herein. All piping and fittings shall withstand an internal working pressure of 1.5 times the shutoff pressure at rated speed without leaking or showing permanent deformation.

3.11.3 Pump accessories. The pump shall be furnished with 6-inch single grooved connections, suction and discharge that accept a coupling conforming to MIL-C-10387. A coupling conforming to MIL-C-10387 and a blind adapter shall be furnished for both the suction and discharge connection to act as cover to prevent entry of foreign material.

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3.12 Engine. The engine shall be an industrial, 2- or 4-stroke cycle, liquid- or air-cooled, diesel engine selected in accordance with MIL-STD-1410, class II. The engine shall deliver power at its continuous duty rating to meet the pump requirements of 3.11. The engine shall be equipped with but not limited to the following:

a. A 24-volt electric cranking system with a charging alternator and batteries suitable for -25 °F starting.
b. Extra heavy duty extended life air cleaner, dry type, replaceable and reuseable, cartridge, with restriction indicator.
c. Fuel filters, primary, water separating type; secondary, each with a drainage provision.
d. Manual engine controls to include a waterproof pushbutton starter switch and engine run/stop switch.
e. Automatic engine emergency shutdown controls in the event of engine overspeed, high temperature, or low lubricant oil pressure.
f. Muffler with guard.
g. Drain tube to permit draining the engine oil into a container without soiling other components of the system.
h. Auxiliary charging receptacle conforming to MIL-C-62122 for recharging the batteries from a remote power source without operation of the engine.
i. Cold weather starting aid conforming to MIL-E-52649, type 3.
j. Value to sample engine oil for testing: Sample must be taken from pressure side of engine. Sampling from sump is not acceptable.
k. Instrumentation consisting of lubricating oil pressure gauge, coolant temperature guage, tachometer-hourmeter, battery charging indicator, an air cleaner restriction indicator and fuel gauge.
l. Heavy duty cooling system capable of maintaining the manufactur 's recommended operating temperature when the pumping assembly is performing at rated capacity in the environmental conditions specified in the specification. If water cooled, the radiator shall have provisions to be cleaned from the front using air.
m. Engine fan belt guard that will provide quick access to and replacement of belt.
n. Oil dip stick conforming to SAE J614.

3.13 Automatic pump controller. The pumping assembly shall be equipped with an automatic pump controller which will reduce engine speed to limit high discharge pressure or low suction pressure. The controller shall sense the discharge and begin to act at 150 psi by reducing engine speed. If the pressure continues to increase, the engine speed will be continuously reduced until idle setting is reached. The controller shall also sense the pressure on the suction side of the pump. If the inlet pressure falls below 20 psi the controller shall start to reduce engine speed. If the inlet pressure continues to fall when the pressure reaches 10 psi, the engine speed will be reduced to idle. The controller shall be capable of being turned off so that the engine speed will be regulated by the engine governor and by the hand throttle.

3.13.1 Operator controls. The operator controls portion of the automatic pump controller shall have two controls. One control shall be a two-position switch providing selection of the desired mode of operation, manual or automatic pressure control. The second operator control knob shall be a combination switch

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and rheostat. From fully counterclockwise to fully clockwise the switch/rheostat position and ranges shall be: (1) stop, (2) start range, and (3) speed set/ discharge pressure set range. Fully clockwise, the switch/rheostat knob shall provide maximum engine speed or maximum discharge pressure depending on the position of the two-position rotary switch. The two operator controls shall be mounted on the instrument panel and permanently identified as specified in 3.9.

3.13.2 Hardware. The pressure sensing and speed command components shall be all solid state electronics. They shall operate an electromechanical actuator to precisely position the fuel control, on the engine, for isochronous (zero droop) speed setting. Military type electrical connectors shall be used for the cables that connect to the automatic pump control modules.

3.13.3 Pressure lines. The pressure lines that provide suction and discharge pressure to the respective transducers in the automatic pump control shall have shutoff cocks. The suction line and associated transducer shall be capable of withstanding at least 150 psig hydrostatic pressure. The discharge line and associated transducer shall be capable of withstanding at least 1.5 times the shutoff pressure at rated speed. The tranducers shall be located to allow for complete drainage to prevent damage during cold storage. The drainage shall be automatic and require no operator action.

3.14 Control and instrument panel. The instrument panel shall be covered on the top and three sides and have a hinged cover which is self supporting and automatically engages over the face of the panel to cover the fourth side. The instrument panel shall conform to the provisions of 3.9 and include as a minimum the following:

a. A helical bourdon tube type compound pressure gauge indicating a vacuum reading from 0 to 30 inches of mercury and from 0 to 200 pounds per square inch positive pressure with provisions for draining.
b. A helical bourdon tube type discharge pressure gauge to indicate a pressure reading from 0 to 200 psi with provision for draining.
c. Tachometer, 0 to 3000 rpm, either mechanical or electrical (may also contain hourmeter).
d. Adjustment screws shall have wrenching recesses or slot filled with a plastic material to prevent indiscriminate adjustment.
e. Voltmeter conforming to MS24532.
f. Engine oil pressure gauge.
g. Engine low oil pressure shutdown with manual reset.
h. Starter switch.
i. Run and stop switch for engine.
j. Low oil pressure shutdown bypass to permit starting.
k. Hourmeter (if not furnished with tachometer).
l. Fuel gauge graduated from empty to full.
m. Instruction plate with procedures for operating engine in manual, normal and alarm modes.
n. Panel shall be illuminated. Illumination shall be in accordance with 3.9 for a protected asset.
o. Air filter restriction indicator.

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3.14.1 Wiring connections. All wiring connections shall use ring terminals, on screw posts with nuts, and lockwashers. All electrical terminals outside the instrument panel shall have standoff metal covers to shield against damage, inadvertent shorting, and weather. All battery and starting system connections shall be equipped with rubber boots. The positive battery terminal cable shall be red and labeled with a "+", and the negative battery cable shall be black and labeled with a "-". All wiring shall be gathered and securely clamped to prevent excessive movement. Military type electrical connectors shall be used wherever possible.

3.15 Fuel system. The fuel tank shall have a sufficient capacity to provide 4-hours operation without refueling. Two 3-way fuel selector valves with corrosion-resisting faceplates with markings of "Off", "Unit Tank", and "Auxiliary" and two 10-foot long fuel hoses conforming to MIL-H-8794 shall be furnished to permit engine operation from the fuel tank or an auxiliary fuel service. The valve shall clearly indicate its position so there is no confusion as to what fuel source is in use. The fuel tank shall be equipped with a filler neck conforming to MS35644, a tank cap conforming to MS35645, and a gasket conforming to MS35643. The bottom surface of the fuel tank shall be protected by a metal guard.

3.16 Wheeled mounting. The chassis shall have two single wheels on an axle assembly attached through leaf springs to the underside of the pump frame. The axle and chassis assembly shall have a rated capacity exceeding the maximum payload, but not less than 5200 pounds and be equipped with a minimum of two shock absorbers for stability.

3.16.1 Tires and wheels. Tires and wheels shall be in accordance with the guidelines of the Tire and Rim Association and shall be of sufficient size and durability to support loads in off-road conditions with no more than 50 psi air pressure. The tires shall contain tubes. Tire pressure shall be indicated directly above the tires "T.P. 50 PSI" in 1-inch high letters. Tires furnished shall not be more than 12 months old on the date of shipment and shall be for all terrain conditions.

3.16.2 Brakes. Brakes shall be of the hydraulic inertia type. Each pumping assembly shall be equipped with a parking hand brake with snap-over-center locking design. The force required to apply the parking brake to keep the trailer from moving down a 25 percent slope shall not exceed 35 pounds. The parking brake handle shall be attached to the side of the tongue and be accessible from the curb side of the towed pump.

3.16.3 Towing extension (tongue). The tongue shall have a nominal height of 30 inches from the ground when the unit is level and parked on a level surface. The tongue shall have a towing eye in accordance with SAE J847. Weight on the towing eye with the pumping assembly parked on a level surface shall be no more than 15 percent and no less than 10 percent of the gross weight of the unit. The tongue shall be equipped with an adjustable, hand-crank style jack incorporated into a tripod design landing gear to permit operation of the jack by one person. The landing gear shall be equipped with a sand shoe with a minimum area of 150 square inches.

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3.16.4 Stabilizing legs. The pumping assembly shall have two heavy duty hand-crank style stabilizing legs, one on each rear corner, with a sand shoe with a minimum of 150-square inch area to permit operation without tipping over when not attached to a towing vehicle. When not in use, the stabilizing leg shall swivel inward to a horizontal position and lock. When in the stored position, the stabilizing leg shall not protrude beyond the widest part of the trailer assembly. The storage and operational position of the stabilizing leg shall maintain the same swivel point. A warning label stating not for use during transport shall be placed over each stabilizing leg.

3.16.5 Lights. The pumping assembly shall have stoplights and tailights in accordance with MS52125. The lights shall contain a switch to allow them to operate in the blackout mode.

3.16.5.1 Electrical connections. The pumping assembly shall have intervehicular connections for the 24-volt electrical system with connectors conforming to MS75020-1

3.17 Pump and engine housing. The pump and engine housing shall be durable and provide easy access to all controls and required maintenance areas. There shall be no protrusions from the basic envelope of the housing. Any external parts, connections or controls must be recessed. All access doors shall be equipped with automatic engaging hold open devices and latches that adequately secure the doors and shall not fail to operate if they suffer from misalignment.

3.17.1 Housing insulation. If noise reducing insulation is used within the housing the insulation shall be protected with a petroleum and heat resisting wrap suitable for the conditions experienced in the housing. All exposed surfaces of the insulation shall be protected by perforated metal covering.

3.18 Lifting and tiedown devices. The pumping assembly shall have lifting and tiedown provisions to secure the pumping assembly to a common carrier and to serve as attachment points to lift the unit by slings. Lifting and tiedowns provisions shall conform to MIL-STD-209 for a type IV item. The pumping assembly shall have markings showing the lift points for lifting by forklift. Lifting and tiedown devices shall be labeled "LIFT HERE" and "TIEDOWN" respectively.

3.19 Transportability. The pump shall be transportable without restrictions by highway, rail, air and marine modes. The applicable transportability criteria are given in MIL-HDBK-157 and MIL-STD-1366. Complete instructions and diagrams for securing the pumping assembly for each mode of transportation shall be included in the operator's manual.

3.19.1 Cargo aircraft. The pumping assembly shall be capable of being transported by C-130, C-141, and C-5 aircraft, and CH-47 (internal and external) and UH-60 (external) helicopters. The pumping assembly shall conform to and meet all restraint criteria specified in MIL-A-8421.

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3.19.2 Rail flatcar. The pump assembly shall withstand the rail impact test of MIL-STD-810, using standard draft gear flatcars.

3.20 Identification, instruction and shipping plates.

3.20.1 Identification plate. Each pumping assembly shall be identified in accordance with MIL-STD-130, on plates made of material conforming to MIL-P-514, type III, composition C, grade B, class 1. The plates shall be attached to the pumping assembly by screws, rivets, or bolts in a place where the plate will be visible and legible but not interfere with the operation and maintenance of the pumping assembly.

3.20.2 Identification of pump inlet and outlet. The pump suction and discharge shall be identified by markings integrally cast in the pump.

3.20.2.1 Identification of direction of flow. The inlet and outlet of the pump casing shall be marked with arrows denoting the direction of flow through the pump. The arrows shall be painted on and shall be black.

3.20.3 Instruction plates. In addition to labels required, each pump shall be equipped with instruction plates, including diagrams, warnings, cautions and operating instructions describing any special or important procedures to be followed in assembling, operating, or servicing the pumping assembly. The pumping assembly shall have an instruction plate to inform the operator on the basic instructions on using the equipment in both the automatic and manual modes. Instrument panel shall have a label which states "Drinking Water Only". One side of the plate shall have instructions for the automatic mode, the other side for the manual mode. Plates should be made of a material conforming to MIL-P-514, type III, composition C, grade B, class 1. Each plate shall be attached to the pumping assembly by screws, rivets, or bolts in a place where the plate will be visible and legible, but not interfere with the operation and maintenance of the pumping assembly. Instruction plates shall conform with the requirements of 3.9.

3.20.4 Performance curve plate. A performance curve plate, 8 x 10-1/2 inches ±1/16 inch, shall be attached to the pumping assembly by screws, rivets, or bolts in a location where the plate will be visible and legible. The chart shall depict the performance curve for the speed required to meet the condition point as well as the condition point itself. The legend shall list psi for liquids of 1.0 specific gravity. The plate material shall be as specified in 3.20.3.

3.20.5 Shipping data plate. Each pump set shall be provided with an aircraft loading data plate conforming to figure 3 of MIL-P-15024/9, type H, style III. The plate shall be attached to the pump and engine assembly as specified in MIL-P-15024 by screws, rivets, or bolts where the plate will be visible and legible.

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3.21 Lubrication. The pumping assembly shall operate as specified herein when serviced with the following military lubricants, grades as applicable to the operating conditions:

Lubricating oil:	MIL-L-2104 or MIL-L-46167
Grease:	MIL-G-10924 or MIL-G-23827

The lubricating points shall be easily visible and accessible. All parts requiring lubrication shall be lubricated before delivery and shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricant used.

3.21.1 Lubrication fittings. The lubrication fittings shall be accessible for a hand-operated grease gun conforming to MIL-G-3859 with a 10-inch flexible extension. Accessiblility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility to lubrication fittings.

3.21.2 Pressure-release device. A pressure release device shall be provided where use of pressure lubricating equipment will damage grease seals.

3.22 Cleaning, treatment, and painting. The portions of the pumping assembly normally painted internally and externally shall be cleaned, treated and painted in accordance with MIL-T-704, type F or G, as applicable. The external surfaces of the pump shall contain no reflective surfaces other than information plates. Unless otherwise specified (see 6.2), the top coat color shall be camouflage green 383 conforming to MIL-C-46168. When specified (see 6.2.1), the pumping assembly shall then be overcoated in accordance with the Government furnished camouflage patterns with MIL-C-46168 topcoat of the colors specified in the camouflage patterns. The first article may be painted camouflage green 383.

3.22.1 Camouflage pattern data. When specified (see 6.2.1), the contractor shall provide camouflage pattern data.

3.23 Battery box. The battery box shall be weatherproofed and made of 304 stainless steel not thinner than 11 gage and shall be designed to rigidly restrain two 12-volt batteries as specified by MS35000. The battery box shall be provided with a lid securely fastened to the box and designed to be easily removed for access to and observation of the battery cells and terminals. The box shall also provide gas venting of the battery cells and contain two holes for drainage. The drain holes shall be positioned so that they drain directly to the ground. Electrolyte as required shall be provided. The electrolyte shall be packaged as required for aircraft transport. A pictorial instruction plate showing battery connections shall be provided on the interior of the box lid.

3.24 Electromagnetic interference. Unless otherwise specified (see 6.2), the pumping assembly shall be equipped for the suppression of electromagnetic interference in accordance with MIL-STD-461 for engine-driven equipment.

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3.25 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or II, as applicable (see 3.25.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish.

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.25.1 Composition. Composition II shall be used only when local air pollution regulations governing the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 442, South Coast Air Quality Management District.

3.26 Workmanship.

3.26.1 Fabrication. Metal used in the fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to ensure uniformity of size and shape. External surfaces shall be free of burrs, sharp edges and corners except where edges or corners are required or where they are not detrimental to safety.

3.26.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Weldments, including weld beads, shall be of homogeneous appearance and form without spattering or other irregularities, complying with the size, configuration and other dimensional requirements to develop the full strength of the parts joined by the welds. No cracks of any type are permitted. All welding is to be produced by either AWS or ASME code qualified welders. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings. Welds shall be in accordance with AWS D1.1.

3.26.3 Welders. Before assigning any welder or welding operator to manual welding work, the contractor shall make available to the contracting officer with certification that the welder or welding operator has passed qualification tests as prescribed for the type of welding to be performed and that such qualification is effective as defined by the particular code of:

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AWS D1.1 - Structural Welding Code - Steel
ASME Boiler and Pressure Vesel Code, Section IX, Welding and Brazing Qualifications.

The government reserves the right to require recertification to any welder or welding operators.

3.26.4 Machine work. Tolerance and gages for metal fit shall conform to commercial industry standards.

3.27 Government furnished property. When specified (see 6.2.1), the following property in the quantity indicated will be furnished by the Government (see 6.5):

Item No.	Description	Quantity
1	Camouflage paint pattern	As required

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.3 Acceptability criteria. Pumping assemblies which conform to all requirements in sections 3 and 5 and pass all applicable examinations and tests in section 4 of this document will be considered acceptable by the Government.

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4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3 and 4.8.1).
b. Quality conformance inspection (see 4.4).
c. Inspection comparision (see 4.7).
d. Inspection of packaging (see 4.8).

4.3 First article inspection. First article inspection shall be performed on one complete unit when a first article sample is required (see 3.2 and 6.2). This inspection shall include the examination of 4.5 and tests of 4.6.1 and 4.6.2.

4.4 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.5 and the tests of 4.6.1.

4.5 Examination. Each pumping assembly shall be examined as specified herein for the following defects. Presence of one or more defects shall be cause for rejection.

Major

101. Materials not as specified.
102. Materials are not corrosion resistant or treated to be corrosion resistant for the applicable storage and operating environment.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes, or treatments.
105. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the pumps.
106. Parts or components missing or not functioning.
107. Assembly incorrect or not complete.
108. Threads not conforming to FED-STD-H28.
109. Pump case and volute not as specified.
110. Impeller not as specified.
111. Seals not as specified.
112. Priming port not as specified.
113. Piping and fittings not as specified.
114. Pump accessories not as specified. Each incorrect accessory shall constitute one defect.
115. Engine not as specified.
116. Automatic pump controller not as specified.
117. Operator controls not as specified.
118. Hardware not as specified.
119. Pressure lines not as specified.
120. Instrument panel not as specified.
121. Wiring connections not as specified.
122. Fuel system not as specified.
123. Wheel mounting not as specified.
124. Tires and wheels not as specified.
125. Brakes not as specified.

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126. Towing extension not as specified.
127. Stabilizing legs not as specified.
128. Lights not as specified.
129. Electrical connections not as specified.
130. Lifting and tiedown devices not as specified.
131. Transportability not as specified.
132. Identification plates not as specified.
133. Instruction plates not as specified.
134. Performance curve plate not as specified.
135. Shipping data plate not as specified.
136. Lubrication not as specified.
137. Lubrication fittings not as specified.
138. Pressure-release device not as specified.
139. Treatment and painting not as specified.
140. Color not as specified.
141. Camouflage pattern data not as specified.
142. Battery box not as specified.
143. Battery not as specified.
144. Fungus and moisture treatment not as specified.
145. Workmanship not as specified.
146. Welder certification not made available to contracting officer.
147. Engine/pump housing not as specified.
148. Insulation material not protected as specified.
149. Materials conforming to 21CFR177 and compounds listed in section 177.1020, 177.1030, 177.1040, 177.1050, 177.1480 of 21CFR not used as specified.

4.6 Tests. When a first article test is required, the first article shall be subjected to the tests specified in 4.6.1 and 4.6.2. Failure to pass any phase of the tests shall be cause for rejection.

4.6.1 Hydrostatic. The pump and fittings shall be subjected to a hydrostatic test pressure of 1.5 times the shut off pressure at rated speed, for a period of 15 minutes, with both suction and discharge valves closed for a period of not less than 5 minutes. Any leakage, cracking, or permanent deformation of the assembly shall constitute failure of the test.

4.6.2 Performance.

4.6.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed in accordance with the applicable provisions of the test code for centrifugal pumps of the standards of the Hydraulic Institute. All performance data shall be corrected to standard sea level conditions of a barometric pressure of 29.92 inches of mercury and water at 60 °F having a specific gravity of 1.00. Water at a temperature of between 60 and 80 °F may be used as the test fluid.

4.6.2.2 Pumping. The pump shall be operated as specified herein to determine pump brake horsepower required, pump efficiency, and the net positive suction head required, based on the data obtained. The resultant data shall be used to plot the corrected performance characteristics.

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4.6.2.3 Priming test. The assembly shall be tested on 10-foot static suction lift and shall fully prime and deliver rated capacity within 2 minutes or less using a 6 inch diameter suction hose. Failure to prime and deliver rated flow in 2 minutes shall constitute failure of the test.

4.6.2.4 Operational test and functional test. The pumping assembly shall be operated for 4 hours at the rated conditions specified in 3.11.1. The pumping assembly shall be operated as required during this period to verify the functional operation of the controls, and the pump shall be examined during the operation for leakage through the pump casing or the shaft seals and for malfunction of any component. Any leakage attributable to defects in design, workmanship, materials, or to the malfunction of any component, or inability of the pump to deliver the minimum capacities specified herein shall constitute failure of this test.

4.6.2.5 Tilted position. The pumping assembly shall be operated for not less than 30 minutes while it is positioned 15 degrees from level in four directions. Evidence of malfunction or misalignment of components shall constitute failure of this test.

4.6.2.6 Safety. The pumping assembly shall be evaluated for safety requirements throughout testing as specified in 4.6.2. Nonconformance to 3.8 shall constitute failure of this test.

4.6.2.7 Human factors engineering inspection. The human factors engineering characteristics requirements shall be inspected during testing. Nonconformance to 3.9 shall constitute failure of this test.

4.6.2.8 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474, 5.1 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum: noise levels shall be measured when equipment is operating under full load; MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 12 equal (horizontal) arc increments; one increment shall include data from the noisiest position; additionally, the noise level at the operator's typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 shall constitute failure of this test.

4.6.2.9 Electromagnetic interference. The first article shall be tested to determine conformance with 3.24. Nonconformance to 3.24 shall constitute failure of this test.

4.6.2.10 Slinging provisions. With the pumping assembly anchored by mean other than the lifting provision being tested, apply the static loads specified in MIL-STD-209 to each provision. Record the angle, maximum load, and the dimensions and exact location of the provision. Weld failure, permanent deformation of the pumping assembly or its lifting provisions, breakage of the provision, or nonconformance to MIL-STD-209 shall constitute failure of this test.

4.6.2.11 Tiedown provisions. With the pumping assembly anchored by means other than the tiedown being tested, apply the static loads specified in MIL-STD-209 to each provision. Record the maximum load and time duration of the

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applied load, and the dimensions and location of the provisions. Weld failure, permanent deformation of the pumping assembly or its tiedown provisions, breakage of the provisions, or nonconformance to MIL-STD-209 shall constitute failure of this test.

4.6.2.12 Ease of maintenance. The pumping assembly shall be evaluated for maintenance accessibility throughout testing. Nonconformance to 3.5.1 shall constitute failure of this test.

4.6.2.13 Reliability. Using the MTBF specified in 3.5.2, the pumping unit shall be tested for reliability. With the pump operating at the condition of 600 GPM and 350 feet total head, test until an "accept" or "reject" decision is reached, in accordance with figure 1. The θ_1, will be 100 hours and θ_0 will be 200 hours. A failure is defined as any malfunction which cannot be corrected within 30 minutes (all waiting periods are excluded) by adjustment, repair or replacement action, using controls and on-equipment tools or parts available to the using element, and is a malfunction which causes or may cause one or more of the following:

a. Failure to commence operation, cessation of operation, or degradation of performance below specified levels.
b. Damage to the pumping unit by continued operation.
c. Safety hazards to personnel.

Entering the reject region on figure 1 shall constitute failure of this test.

4.6.2.14 Environmental.

a. The pump shall be tested for storage and operation at the temperatures specified in 3.7.1 and 3.7.2, in accordance with procedure I, method 501 and 502 of MIL-STD-810. Operating time shall be a minimum of 3 hours for each test.
b. Failure of the pump to operate during or after the required tests shall be cause for rejection of the pump.

4.6.2.15 Rail impact. The first article pumping assembly shall be tested in accordance with MIL-STD-810 rail impact test to show conformance to transportability requirements of 3.19.2, any permanent deformation or failure, or inoperability after test shall constitute failure.

4.7 Inspection comparison. The Government may select pumping assemblies any time during the contract production period and subject these pumping assemblies to the examination specified in 4.5. The inspection will be performed by the Government, at a site selected by the Government, on pumping assemblies selected at random from those which have been accepted by the Government and will not include any previously inspected pumping assemblies. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effects as failure of those tests specified as inspection comparison.

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4.7.1 Inspection failure. Failure of an inspection comparison pumping assembly to meet any requirement specified herein, during and as a result of the examination and tests specified in 4.6, shall be cause for refusal by the Government to continue acceptance of production pumping assemblies until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on pumping assemblies previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all pumping assemblies accepted prior to the completion of the inspection comparison are similarly deficient, unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 First article inspection of packaging. The first article packaging shall be examined and tested in accordance with the first article pack (preproduction pack) inspection provisions of MIL-P-10603, and for the additional examination defects specified in 4.8.2.1 herein.

4.8.2 Quality conformance inspection. Packaging of pump assemblies shall be examined in accordance with the quality conformance inspection provisions of MIL-P-10603, and for additional examination defects specified in 4.8.2.1 herein.

4.8.2.1 Examination. Sample packs, selected in accordance with the sampling requirements of MIL-P-10603, shall be examined for the following defects, in addition to the defects lised in MIL-P-10603. The AQL shall be as specified therein.

Major	Defect	
150.	Cooling system of liquid cooled engines not preserved as specified.	5.2.1/5.2.2
151.	Commercial packaging not as specified.	5.2.2
152.	Pump preservative not as specified.	5.2.1

5. PACKAGING

5.1 First article packaging. Unless otherwise specified (see 6.2), the contractor shall furnish a first article pack for examination and testing within the time frame specified (see 6.2), to prove prior to the start of production packaging, that the applied preservation, packing, and marking comply with the requirements of this specification. Examination and testing shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The first article pack may utilize either the first article model or a production model. When the first article model is utilized, and the Government requests a comparison between the first article model and a production model, any preservation and packing shall be removed by the contractor at no expense to the Government.

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5.2 Packaging. Packaging of the pumping assemblies shall be level A, B, or commercial as specified (see 6.2). Preservation materials and their application shall not adversely affect the drinkability of water that may be pumped through this assembly.

5.2.1 Level A and B. Levels A and B packaging, which includes preservation, packing and marking, shall be in accordance with the applicable provisions for levels A and B of MIl-P-10603 except that type P-14 preservative shall be used in lieu of type P-10 and the cooling system of a liquid cooled engine shall be preserved by the water and antifreeze procedure specified in MIL-E-10062 using antifreeze conforming to MIL-A-46153. Alternatively, when specified (see 6.2), the pumping assemblies may be packed in crates or boxes as specified in MIL-P-10603 for units mounted on nonhighway type trailers.

5.2.2 Commercial. Commercial packaging, which includes preservation, packing and marking, shall be in accordance with ASTM D 3951 except that the cooling system of a liquid cooled engine shall be preserved as specified in 5.2.1. Additionally, each shipping container, as applicable, shall be marked with the gross weight and cube.

5.3 Depreservation guide. Two depreservation guides shall be prepared. One guide shall be placed in a waterproof envelope marked "Depreservation Guide" and shall be secured to the pump assembly in a conspicuous location. The second guide shall be preserved with the technical publications and the bag shall be marked to indicate that the depreservation guide is inside.

6. NOTES

6.1 Intended use. The pumping assembly is intended to pump potable water through a 6 inch hoseline in cold and hot temperatures, and is a component of the Tactical Water Distribution System. Operating conditions could be dry or humid with frequent periods of blowing sand.

6.2 Ordering data.

6.2.1 Acquistion requirements. Acquistion documents should specify the following:

a. Title, number, and date of this purchase description.
b. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
c. When a first article is not required for first article inspection and approval (see 3.2, 4.3 and 6.3).
d. Color of finish coat if not as specified (see 3.22).
e. When electromagnetic interference suppression is not required (see 3.24).
f. Time frame required for submission of the first article pack (see 5.1).
g. Time frame when camouflage pattern data is required (see 3.22.1).
h. When first article packaging is not required (see 5.1).
i. Degree of packaging required (see 5.2).
j. When pumping assembly is to be packed in crate or box (see 5.2.1).

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6.2.2 Data requirements. When this specification is used for an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data requirements	Applicable DID No.
a. 3.22.1	Color photograph prints	DI-MISC-80192
b. 3.22.1	Camouflage line art data	DI-MISC-80176

6.3 First article. When a first article inspection is required, the items should be a preproduction model. The first article should be one or more complete units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the first articles.

6.4 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack wil not relieve the contractor of his obligation to preserve, pack, and mark the pumping assembly in accordance with this specification.

6.5 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.27.

6.6 Depreservation guide. The contracting officer should arrange to furnish the necessary copies of DA Form 2258, when requested by the contractor (see 5.3).

6.7 Subject term (key word) listing.

Hoseline, pump
Pumping assembly, centrifugal
Water, portable, movement

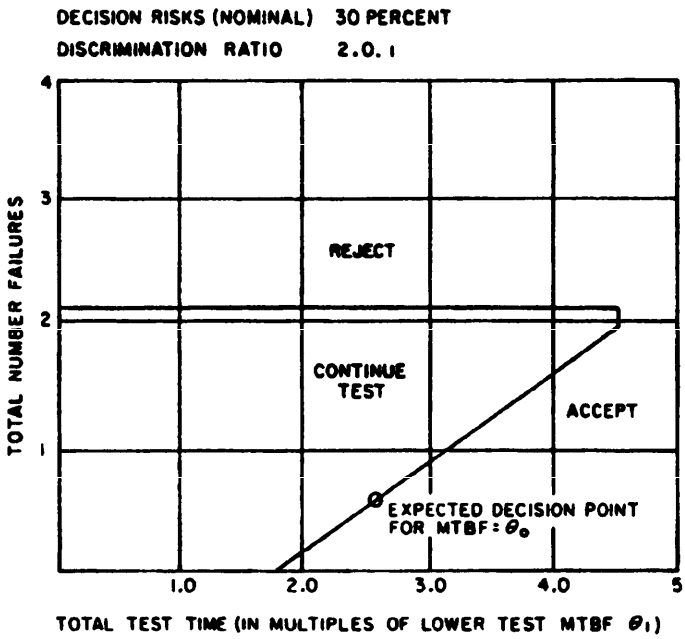
6.8 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 4320-A263

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TOTAL TEST TIME *

NUMBER OF FAILURES	REJECT (EQUAL OR LESS)	ACCEPT (EQUAL OR MORE)
0	N/A	1.72
1	N/A	3.10
2	N/A	4.50
3	4.5	N/A

* TOTAL TEST TIME IS TOTAL UNIT HOURS OF EQUIPMENT ON TIME AND IS EXPRESSED IN MULTIPLES OF THE LOWER TEST MTBF. REFER TO 4.6.2.13 FOR MINIMUM TEST TIME PER EQUIPMENT.

FIGURE 1. Accept-reject criteria.

X-4281A

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-53054A(ME)	Pumping Assembly, Water, Wheel Mounted, DED, Self Priming Centrifugal, 6-Inch, 600 GPM at 350 Feet Total Head

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

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